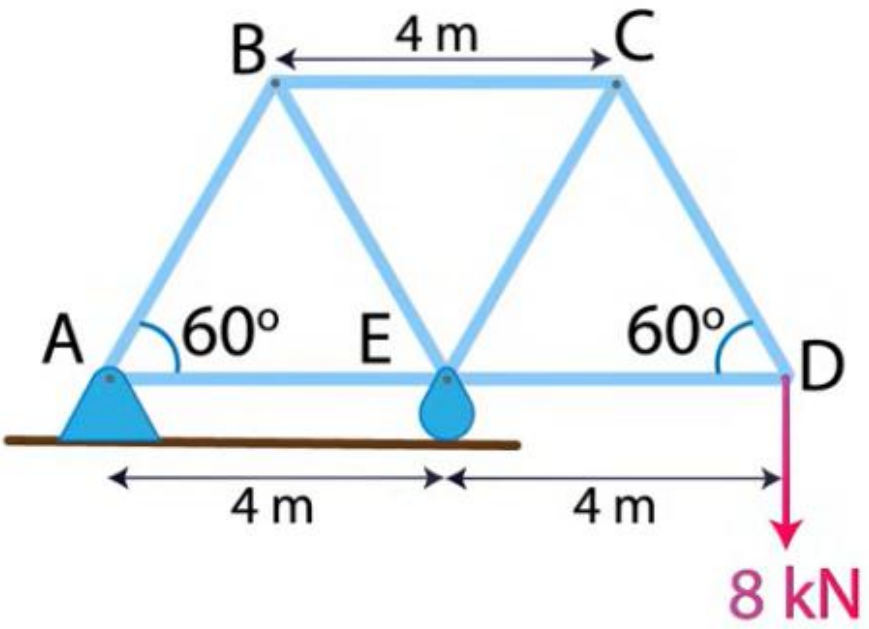
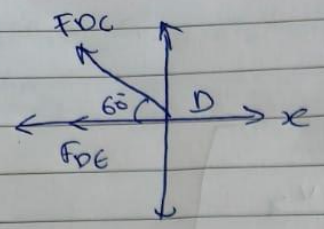
Case Study 1

Truss Joint Force Analysis

Assignment 1: Assemble equilibrium equations for a small 2D truss and solve member forces using simultaneous linear equations



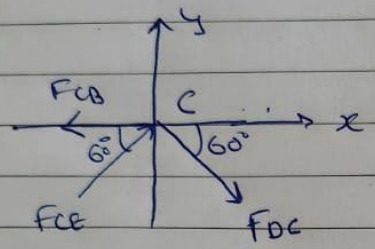


$\uparrow^{+} \Sigma F_y = 0$

$-8 + F_{DC} \sin 60° = 0$

$\overset{+}{\leftarrow} \Sigma F_x = 0$

$F_{DE} + F_{DC} \cos 60° = 0$

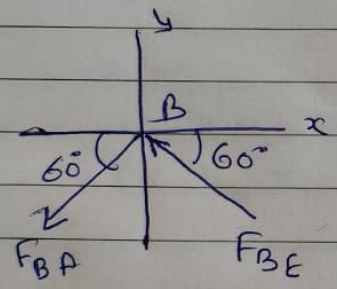


$\uparrow^{+} \Sigma F_y = 0$

$F_{CE} \sin 60° - F_{DC} \sin 60° = 0$

$\rightarrow^{+} \Sigma F_x = 0$

$F_{DC} \cos 60° - F_{CB} + F_{CE} \cos 60° = 0$

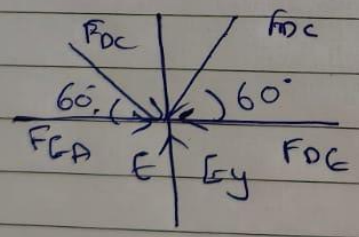


$\uparrow^{+} \Sigma y = 0$

$F_{BE} \sin 60° - F_{BA} \sin 60° = 0$

$\overset{+}{\rightarrow} \Sigma F_x = 0$

$F_{DC} - F_{BE} \cos 60° - F_{BA} \cos 60° = 0$

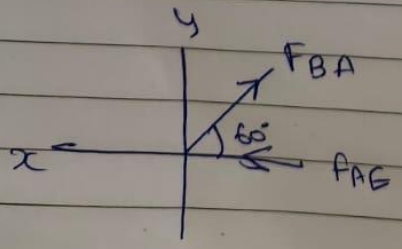


$\uparrow^{+} \Sigma F_y = 0$

$E_y - F_{DC} \sin 60° - F_{DC} \sin 60° = 0$

$\overset{+}{\rightarrow} \Sigma F_x = 0$

$F_{EA} - F_{DE} - F_{DC} \cos 60° + F_{DC} \cos 60° = 0$



$\Sigma F_y = 0$

$A_y + F_{AB} \sin 60° = 0$

$\Sigma F_x = 0$

$A_x + F_{AB} \cos 60° + F_{AE} = 0$

Matlab Code

/MATLAB Drive/untitled8.m

```
clc;
clear;
syms FAB FBC FCD FAE FED FBE FCE Ax Ay Ey
c60 = cosd(60);
s60 = sind(60);
%% Joint A
eq1 = Ax + FAB*c60 + FAE == 0;
eq2 = Ay + FAB*s60 == 0;
%% Joint B
eq3 = -FAB*c60 + FBC + FBE*(2/4) == 0;
eq4 = -FAB*s60 - FBE*(3.464/4) == 0;
%% Joint C
eq5 = -FBC + FCD*c60 - FCE*(2/4) == 0;
eq6 = FCD*s60 - FCE*(3.464/4) == 0;
%% Joint D
eq7 = -FCD*c60 - FED == 0;
eq8 = -FCD*s60 - 8 == 0;
%% Joint E
eq9  = Ey - FBE*(3.464/4) - FCE*(3.464/4) == 0;
eq10 = FAE + FED + FBE*(2/4) - FCE*(2/4) == 0;

sol = solve([eq1 eq2 eq3 eq4 eq5 eq6 eq7 eq8 eq9 eq10], ...
            [FAB FBC FCD FAE FED FBE FCE Ax Ay Ey]);
disp('Member Forces (kN):')
fprintf('FAB = %.2f\n', double(sol.FAB))
fprintf('FBC = %.2f\n', double(sol.FBC))
fprintf('FCD = %.2f\n', double(sol.FCD))
fprintf('FAE = %.2f\n', double(sol.FAE))
fprintf('FED = %.2f\n', double(sol.FED))
fprintf('FBE = %.2f\n', double(sol.FBE))
fprintf('FCE = %.2f\n', double(sol.FCE))

disp('Support Reactions:')
fprintf('Ax = %.2f kN\n', double(sol.Ax))
fprintf('Ay = %.2f kN\n', double(sol.Ay))
fprintf('Ey = %.2f kN\n', double(sol.Ey))
```

Result

```
Command Window
Member Forces (kN):
FAB = 0.00
FBC = 0.00
FCD = -9.24
FAE = -9.24
FED = 4.62
FBE = -0.00
FCE = -9.24
Support Reactions:
Ax = 9.24 kN
Ay = -0.00 kN
Ey = -8.00 kN
Press Ctrl + Shift + P to generate code with Copilot
```